

XPCyber Challenge Group Analyst Debrief Report

Week #8 XPCyber Challenge - Analyzing Attack Artifacts

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Term 2

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Executive Summary

This investigation was conducted in response to a suspected compromise of the production web server within the XPCyber environment. Initial context from system administrators indicated unusual activity and a potential attack, prompting a forensic analysis of the provided system image (var.img). The primary objective was to identify indicators of compromise while maintaining strict evidence integrity.

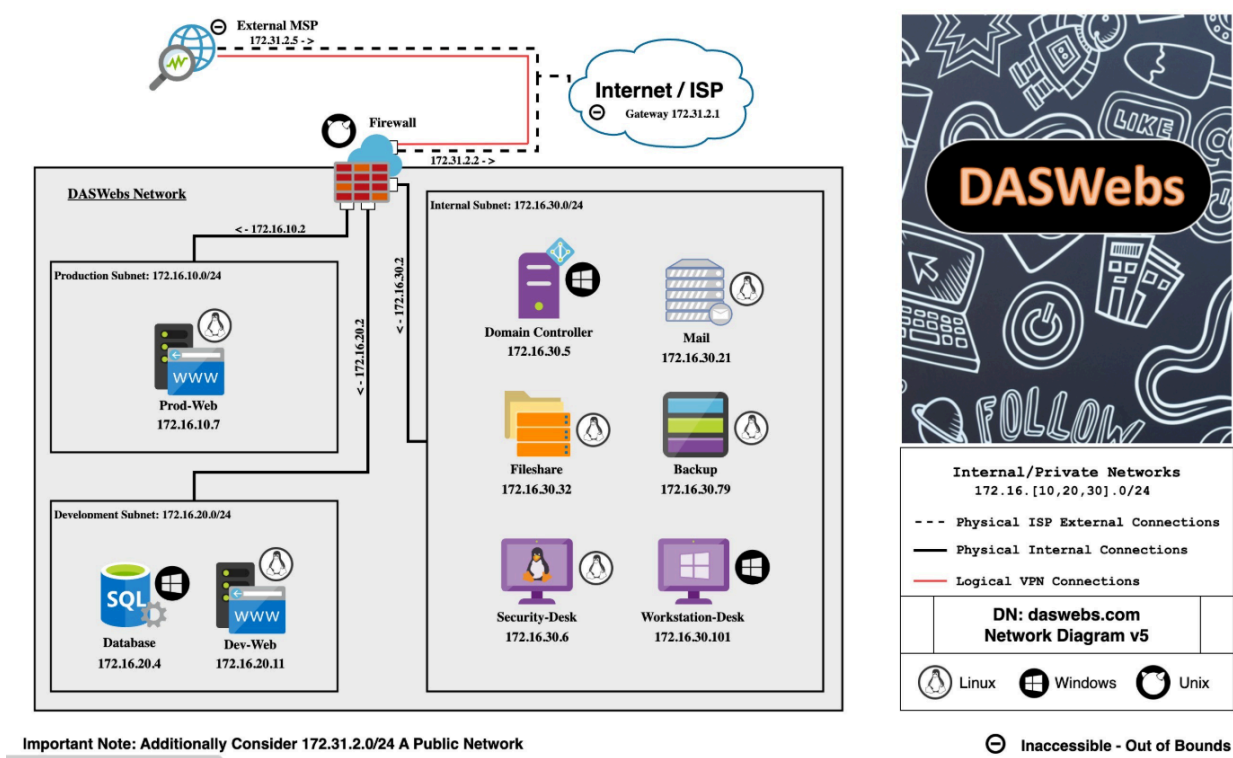
We as analysts approached the analysis by mounting the image in a read-only state to prevent modification and systematically examining file structures. After initial review yielded limited findings, elevated privileges were used to access restricted directories, with a focused shift toward system logs as the most likely source of attack artifacts. Analysis of the access_log and error_log files revealed suspicious activity originating from an external IP address (172.31.2.195), including repeated POST requests to /xmlrpc.php associated with WPScan, a known WordPress vulnerability scanning tool. Further evidence showed probing attempts targeting sensitive configuration files such as wp-config.php, indicating reconnaissance and potential exploitation attempts.

Based on these findings, it was concluded that the system experienced a targeted reconnaissance and attempted exploitation of WordPress vulnerabilities. The most critical evidence was preserved and extracted to the designated evidence directory using forensically sound methods. It is recommended that the organization disable or restrict access to xmlrpc.php, implement web application firewall protections, monitor for abnormal external IP activity, and ensure timely patching of WordPress components to mitigate similar threats in the future.

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Technical Steps & Analysis

Network Map



Chat logs

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Gary Thatcher @gthatcher | *Sr. Systems Admin*

Our production web server looks like it's been attacked, but I don't have many details on what actually occurred. As soon as I noticed a potential attack, I shut down Prod-Web and imaged the /var partition. I placed the .img on Security-Desk's Desktop.



Ione Leventis @ileventis | *Security Analyst*

You really shouldn't just shut down machines that you think have been potentially compromised; reach out to me regarding what's going on first.



Gary Thatcher @gthatcher | *Sr. Systems Admin*

Understood @ileventis. Will ask you to advise going forward.



Ione Leventis @ileventis | *Security Analyst*

@playerone, given this has already occurred, please perform an investigation, extracting any evidence you find.



Ione Leventis @ileventis | *Security Analyst*

Please be aware that it's very important you don't alter in any way the evidence you're provided. The checksum of the image file provided must be the same as when @gthatcher imaged it, and be sure to leave it on the Desktop of Security-Desk. In addition, we're asking you to limit your evidence gathered to five files, as we pay per file to have them analyzed by a forensic expert.



Richard LeGrand @rlegrand | *CEO*

HAVE WE BEEN HACKED?!?!? HOW COULD THIS HAPPEN TO MY COMPANY?? @playerone PERFORM THIS ANALYSIS ASAP, ASAP!!



Ione Leventis @ileventis | *Security Analyst*

Uhhh... let's not jump to conclusions yet. @playerone, the image file is named 'var.img', and it's on Security-Desk's Desktop, as @gthatcher stated. Feel free to use whatever tools you need to analyze and extract the evidence you find without modifying it. Place any files you find in the 'evidence' directory, which is also on the Desktop of Security-Desk. Be sure to avoid placing evidence in any subdirectories within that directory.



Gary Thatcher @gthatcher | *Sr. Systems Admin*

Thank you, @playerone! Best of luck.

Notes

- Up to 5 files
- Not always visible
- Verify hash

Approach steps and actions taken

We started by opening the var.img which created a read only version to allow us to examine it without altering it in any way. We went through the entire file and checked every file that we could open to look for evidence of the attack. When we found no evidence we decided to utilize the root user to view the rest of the files using the elevated permissions. From there we stayed in the log folder as we knew this is where the information would likely be located. We switched to the httpd folder and performed the list function to see all the files that were in the folder.

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```
(root@security-desk)-[/media/playerone/e9cfd082-009c-4dbc-8359-a28f16446d9c/log]
# cd httpd

(root@security-desk)-[/media/playerone/e9cfd082-009c-4dbc-8359-a28f16446d9c/log/httpd]
# ls
access_log  error_log

(root@security-desk)-[/media/playerone/e9cfd082-009c-4dbc-8359-a28f16446d9c/log/httpd]
# cat access_log | tail -50
172.31.2.195 - - [18/Oct/2022:19:16:19 +0000] "HEAD /wp-config.php1 HTTP/1.1"
404 - "http://172.16.10.7" "WPScan v3.8.20 (https://wpscan.com/wordpress-security-scanner)"
172.31.2.195 - - [18/Oct/2022:19:16:19 +0000] "HEAD /wp-config.phpbak HTTP/1.1"
404 - "http://172.16.10.7" "WPScan v3.8.20 (https://wpscan.com/wordpress-security-scanner)"
172.31.2.195 - - [18/Oct/2022:19:16:19 +0000] "HEAD /wp-config.phpb HTTP/1.1"
404 - "http://172.16.10.7" "WPScan v3.8.20 (https://wpscan.com/wordpress-security-scanner)"
172.31.2.195 - - [18/Oct/2022:19:16:19 +0000] "HEAD /wp-config.phpa HTTP/1.1"
404 - "http://172.16.10.7" "WPScan v3.8.20 (https://wpscan.com/wordpress-security-scanner)"
172.31.2.195 - - [18/Oct/2022:19:16:19 +0000] "HEAD /wp-config.phpc HTTP/1.1"
404 - "http://172.16.10.7" "WPScan v3.8.20 (https://wpscan.com/wordpress-security-scanner)"
```

Access log

That listed out the access_log and error_log. We used the cat function next to read the last 50 entries.

```
File Actions Edit View Help
910] script '/var/www/html/searchreplacedb2.php' not found or unable to stat,
referer: http://172.16.10.7
[Tue Oct 18 19:16:12.110056 2022] [:error] [pid 1121] [client 172.31.2.195:49
912] PHP Fatal error: Call to undefined function _deprecated_file() in /var/
www/html/wp-includes/rss-functions.php on line 9, referer: http://172.16.10.7
[Tue Oct 18 19:16:13.292880 2022] [:error] [pid 1118] [client 172.31.2.195:49
914] script '/var/www/html/emergency.php' not found or unable to stat, refere
r: http://172.16.10.7
[Tue Oct 18 19:16:18.269441 2022] [:error] [pid 1397] [client 172.31.2.195:49
928] script '/var/www/html/backup.wp-config.php' not found or unable to stat,
referer: http://172.16.10.7
[Tue Oct 18 19:16:18.275255 2022] [:error] [pid 1396] [client 172.31.2.195:49
920] script '/var/www/html/wp-config copy.php' not found or unable to stat, r
eferer: http://172.16.10.7
[Tue Oct 18 19:16:18.275761 2022] [:error] [pid 1119] [client 172.31.2.195:49
918] script '/var/www/html/wp-config - Copy.php' not found or unable to stat,
referer: http://172.16.10.7
[Tue Oct 18 19:16:18.288104 2022] [:error] [pid 1119] [client 172.31.2.195:49
918] script '/var/www/html/wp-config-backup.php' not found or unable to stat,
referer: http://172.16.10.7
[Tue Oct 18 19:16:18.345592 2022] [:error] [pid 1120] [client 172.31.2.195:49
922] script '/var/www/html/wp-config.local.php' not found or unable to stat,
referer: http://172.16.10.7

(root@security-desk)-[/media/playerone/e9cfd082-009c-4dbc-8359-a28f16446d9c/log/httpd]
#
```

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Error log

The access log showed us POST /xmlrpc.php HTTP/1.1 "WPScan v3.8.20 (Is WordPress XMLRPC a Security Problem?, 2021)" IP: 172.31.2.195. We noted that this was an out of network IP address performing a xmlrpc.php, which is a known WordPress attack surface as it exploits wordpress vulnerabilities. specifically using remote code execution for things like brute force attacks. We then examined the errors log by following the same cat process. In the errors log we noted that the attacker was trying to access sensitive wordpress files like wp-config.php, wp-config-backup.php, wp-configcopy.php, wp-config.local.php. It was going through and probing all of the wordpress files for information. After confirming that these were the files relevant to the potential attack we used the command line to extract and preserve integrity of the files when we moved them to the evidence folder. We used `cp -p access_log /home/playerone/Desktop/evidence/` and `cp -p error_log /home/playerone/Desktop/evidence/`.

```
(root@ security-desk)-[/media/playerone/e9cfd082-009c-4dbc-8359-a28f16446d9c/log/httpd]
# cp -p access_log /home/playerone/Desktop/evidence/
```

```
(root@ security-desk)-[/media/playerone/e9cfd082-009c-4dbc-8359-a28f16446d9c/log/httpd]
# cp -p error_log /home/playerone/Desktop/evidence/
```

At this point we concluded our investigation as we had found all relevant files. *** Issues with what files were needed***

Answers Submitted

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FINAL CHECK DETAILS

- ✓ Check #1: Forensic Image Unaltered [Should Stay Green]
- ✓ Check #2: Unaltered Evidence Obtained Following Specified Protocol

Conclusion and Recommended Actions

The investigation that was conducted by us determined that the production web server was subjected to a targeted reconnaissance and exploitation attempt leveraging known WordPress vulnerabilities. This conclusion is supported by multiple indicators found within the system logs, specifically repeated POST requests to `/xmlrpc.php` associated with WPScan activity and originating from an external IP address (172.31.2.195). Additionally, the error logs show systematic probing of sensitive configuration files such as `wp-config.php` and its variants, which is consistent with enumeration techniques used to identify misconfigurations or exposed credentials. The consistency, repetition, and nature of these requests demonstrate intentional malicious behavior rather than normal web traffic. The integrity of the findings is defensible due to the use of a read-only forensic image and preservation methods (`cp -p`) that ensured metadata and evidence were not altered during extraction.

From a remediation standpoint, the most pragmatic actions focus on reducing the attack surface and strengthening monitoring controls. First, access to `xmlrpc.php` should be disabled or restricted, as it is a well-documented vector for brute force and remote code execution attacks in WordPress environments. Second, the implementation of a Web Application Firewall (WAF) is recommended to detect and block automated scanning tools such as WPScan. Third, all WordPress core files, plugins, and themes should be audited and updated to ensure known

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vulnerabilities are patched. Additionally, log monitoring and alerting should be enhanced to detect abnormal request patterns, particularly repeated external requests targeting sensitive endpoints. Finally, network-level controls such as IP blocking or rate limiting should be applied to mitigate repeated malicious traffic from identified sources.

Three and two debrief

For this lab there were several things that went well as well as ones that need improvement. One thing that went well is we were very thorough when it came to looking at the files in the var.img. We checked everything in the disc image including every file and folder. In hindsight most of what we searched was irrelevant to what we were looking for but we wanted to make sure we didn't miss anything outside of the traditional locations. A second thing that went well is that when we got confused even though we did technically finish the lab we were able to contact our professor, you and set up a meeting to get clarification.

One thing that didn't go well is that the way the checks were set up was confusing. So it counted us as having the files even if they all weren't in the evidence folder at the same time So it said our lab was complete when we in actuality only had two files in there. This is something that we discussed in the meeting so it should hopefully be fixed for future labs that students need to do. Another thing that went well is even though we did forget some screenshots because we didn't follow the correct process we were told that we could just submit it as is and it would be fine so we didn't have to go back through and redo the lab. One final area that could use improvement is that there was some miscommunication on what method we were using to copy the files, one partner used command line and the other partner did not and what we realized is that without the command line the process that the other partner used altered the forensic state of

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the file. This meant that we lost one of the check marks and we were unable to figure out a way to return it to its proper state so we had to end the lab session and start a new one. Overall I think the way he had had more clarification going into the lab it would have gone smoother but all things considered it went fairly well.

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NICE Challenge Project

Challenge Submission Report

<https://portal.nice-challenge.com/reports/verify/E1DE5-52E6-DCD0D/>

Submission ID: 153170

Timestamp: 3/22/2026 12:16 AM UTC

Name: Anna Fanco

Challenge ID: 189

Challenge Title: Analyzing Attack Artifacts



This report has not been published by a curator. The NICE Challenge Project cannot vouch for its accuracy.

Scenario

We have recently become aware of an attack on our production web server. In response, an image has been made of the server's /var partition that needs to be analyzed for evidence of the attack. Any identified evidence must be extracted in a forensically sound manner so it can later be analyzed by an external forensic expert.

Duration

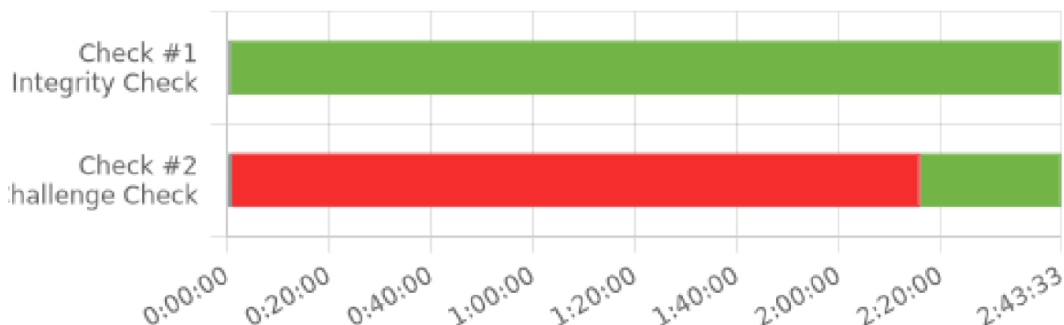
2:43

Final Check Details

- ✓ Check #1: Forensic Image Unaltered [Should Stay Green]
- ✓ Check #2: Unaltered Evidence Obtained Following Specified Protocol

Full Check Pass

Full: 2/2



Element

Cyber Enablers

Work Role

Cyber Crime Investigator

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DCWF Tasks

- 649: Identify, collect, and seize documentary or physical evidence, to include digital media and logs associated with cyber intrusion incidents, investigations, and operations.

Knowledge, Skills, and Abilities

- 217: Skill in preserving evidence integrity according to standard operating procedures or national standards.
- 290: Knowledge of processes for seizing and preserving digital evidence (e.g., chain of custody).
- 369: Skill in collecting, processing, packaging, transporting, and storing electronic evidence to avoid alteration, loss, physical damage, or destruction of data.
- 1158: Knowledge of cybersecurity principles.
- 1159: Knowledge of cyber threats and vulnerabilities.

Centers of Academic Excellence Knowledge Units

- Cybersecurity Foundations
- Digital Forensics
- Host Forensics
- Media Forensics
- Operating Systems Concepts

References

Is WordPress XMLRPC a security problem? (2021, June 29). WPScan.
<https://wpscan.com/blog/is-wordpress-xmlrpc-a-security-problem/>

Kent, K., & Souppaya, M. (2006). *Special Publication 800-92 Guide to Computer Security Log Management Recommendations of the National Institute of Standards and Technology*. NIST.
<https://nvlpubs.nist.gov/nistpubs/Legacy/SP/nistspecialpublication800-92.pdf>

Cichonski, P., Millar, T., Grance, T., & Scarfone, K. (2012). Computer security incident handling guide. *Computer Security Incident Handling Guide*, 2(2).
<https://doi.org/10.6028/nist.sp.800-61r2>

Souppaya, M., & Scarfone, K. (2013). Guide to Malware Incident Prevention and Handling for Desktops and Laptops. *Guide to Malware Incident Prevention and Handling for Desktops and Laptops*, 1(1). <https://doi.org/10.6028/nist.sp.800-83r1>